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Final Report

Learning When *Not* to Trade: Reinforcement Learning for
Cross-Venue Latency Arbitrage After Transaction Costs

Group X | Application Project

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AI-use disclosure: AI coding agents implemented substantial parts of the software and drafted this report under the team's direction. The full statement follows the conclusion.

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Abstract

We asked whether a reinforcement learning agent can trade Bitcoin perpetual futures between Binance and Hyperliquid only when the price gap between the two venues is large enough to pay for fees, spread and execution delay, and stay flat otherwise. We prepared 30 days of recorded order books (3 September to 2 October 2026, UTC) into 8,245 causally aligned five-minute replay episodes and built a Gymnasium environment that fills orders 150 ms after the decision against the recorded Hyperliquid book and charges a taker fee on each fill. Two PPO designs were trained and selected on validation days only. The first, a four-action policy trained for one million steps on three seeds plus a higher-entropy retry, cut its share of entry actions below 1% within about 230,000 steps; the selected policy made zero trades on validation and on the four held-out test days. A fixed rule that enters when the gap reaches 10 basis points (bps; 1 bps is 0.01%), run as a control in the same environment, made 16 test trades for a net \$1.19 with a taker fee of 3.5 bps on each fill. At 4.5 bps per fill, the rate we used later, the same selection procedure would have picked a threshold that never trades. The second design, a skip-or-enter decision at screened candidate gaps with a supervised warm start, also ended with zero validation trades for all three PPO seeds. We report properties of the problem that are consistent with the failure, without testing which of them caused it: gaps of 7 bps or more, roughly one round-trip cost, occurred in under 0.1% of validation and test seconds (0.084% and 0.097%); staying flat returns exactly zero while early trading loses money; and 99.7% of the training seconds with a gap of 7 bps or more fell in a six-day period when the two venues’ prices were offset and did not converge. A weaker offset of the same kind recurred on validation (daily median -2.8 bps on 25 September). A zero-trade policy is abstention, and we report it as a failure to learn the intended behaviour.

1 Introduction

Bitcoin perpetual futures (futures contracts with no expiry date) trade on many venues at the same time, and a price move does not reach every venue’s order book at the same moment. A trader who sees a move first on one venue can buy or sell on a venue whose quotes have not yet adjusted, then exit when the two prices converge. This is latency arbitrage. Budish, Cramton and Shim [1] describe the same mechanism in equity markets, where fast traders profit by trading against quotes that have not yet caught up with a price change elsewhere.

An order book lists resting buy orders (bids) and sell orders (asks) at each price level. The best bid and the best ask form the top of the book, the spread is the best ask minus the best bid, and depth is the size available at the displayed levels. The mid price is the average of the best bid and the best ask. We measure the gap as the Binance mid minus the Hyperliquid mid, divided by the Hyperliquid mid, in basis points (bps; 1 bps is 0.01%). Across the 2,473,500 one-second decision rows we kept, the median absolute gap was 1.29 bps.

A visible gap is not a profit. An order that executes immediately against resting orders is a taker order: it buys at the ask and sells at the bid, so relative to the mid it pays half the spread on each fill and one full spread over a round trip (entry and exit). Each fill also pays a taker fee and can suffer slippage, an execution price worse than the quoted one. Below, “per side” means per fill. With our base assumptions of a 3.5 bps fee and 0.1 bps slippage per side, and a median Hyperliquid spread of about 0.12 bps, a round trip costs $2 \times 3.5 + 2 \times 0.1 + 0.12 \approx 7.3$ bps; at 4.5 bps per side, the fee in our second attempt, it costs about 9.3 bps. On the validation and test days, fewer than 0.1% of decision seconds had an absolute gap of 7 bps or more. A profitable policy has to trade only on the few gaps larger than this cost. The internal feasibility study

that motivated the project reported similar magnitudes: a median gap of about 1 bps against a round-trip cost of about 7 bps [2].

Our research question was whether proximal policy optimisation (PPO) [3] can learn this selective behaviour from recorded order books with all costs in the reward, and beat a fixed threshold rule under identical costs on later, unseen days. Reinforcement learning has been applied to related microstructure problems such as optimal order execution [4]. Here, profitable entries are rare and staying flat returns exactly zero. The proposal rated “RL training is unstable or learns only to abstain” as a medium risk, and abstention is what happened. Two PPO designs, with seven trained PPO policies between them, ended with zero validation trades, and the selected policy of the first design also made zero trades on the held-out test, while a 10 bps threshold control made a small profit there.

The project produced a quality-checked, causally aligned replay dataset with a SHA-256 hash for every episode; a replay environment with delayed fills against the recorded book, fees on both sides, and freshness and risk checks, covered by 41 environment tests and 20 evaluation tests, all passing (the tracked suite has 168 tests); two pre-declared PPO experiments with validation-only selection and a single test evaluation; and an analysis of the failure based on the training logs and the data (Section 6). Section 9 lists the changes we would make next.

2 Data

2.1 Source and research range

The team recorded both venues’ public order-book streams into one ClickHouse table. Each row is one book snapshot with the exchange timestamp, the local receive timestamp, and bid and ask price and size arrays. Binance rows come from the btcusdt@depth20@100ms stream and always have 20 levels per side. Hyperliquid rows come from its 12Book stream and always have 5 levels per side; the proposal had assumed 10 levels for both. The data are public market feeds and contain no personal information. An inventory taken at 16:30 UTC on 3 October 2026 counted 32,888,724 rows (27,464,104 Binance and 5,424,620 Hyperliquid). Capture began on 27 August, but an outage left 30 August almost empty, 31 August and 1 September missing, and 2 September partial. Before any modelling we fixed the research range to the 30 complete days from 3 September to 2 October, which hold 28,917,283 raw rows (24,146,574 Binance and 4,770,709 Hyperliquid). The split is chronological: 22 days for training, 4 for validation and 4 for testing (Table 1). The day of 3 October (780,739 Binance and 158,589 Hyperliquid rows) was prepared later as a separate holdout.

Table 1: Splits and gap statistics on prepared one-second decision rows. Gap is $10^4 \times (\text{Binance mid} - \text{Hyperliquid mid}) / \text{Hyperliquid mid}$. “Dropped” counts five-minute windows rejected by the quality filter.

Split	Dates (2026)	Windows		gap (bps)		Seconds with gap ≥ 7 bps
		(dropped)	Decisions	Median	p99	
Train	3–24 Sep	5,984 (352)	1,795,200	1.34	12.57	18.04%
Validation	25–28 Sep	1,121 (31)	336,300	1.30	5.20	0.084%
Test	29 Sep–2 Oct	1,140 (12)	342,000	1.04	4.86	0.097%
Fresh day	3 Oct	286 (2)	85,800	0.61	2.31	0.0047%

2.2 Causal alignment and quality filtering

Decisions are made on a grid of whole UTC seconds. Each decision second also has two execution rows, at +150 ms and +300 ms, which are used to fill delayed orders, so the replay has three rows per second. Every grid row takes, for each venue, the latest snapshot received at or before its timestamp, so no row uses information that arrived later. Each grid row also records two ages per venue: the receive age, the row's time minus the local receive time of that snapshot, and the quote age, the row's time minus the snapshot's exchange timestamp. A grid row was marked invalid if either venue had no earlier snapshot; if the latest snapshot failed the validity checks (finite positive prices and sizes, strictly ordered levels, a best bid below the best ask); if the receive age exceeded 2,000 ms or the quote age exceeded 3,000 ms; if a spread exceeded 100 bps; or if the gap exceeded 250 bps. Episodes are fixed 300-second UTC windows (900 rows and 300 decisions). A window with any invalid row was dropped whole, so a feed outage never appears inside an episode (Section 8.2 discusses the bias this creates).

Of 8,640 windows, 8,245 (95.43%) were kept. Hyperliquid staleness caused almost every rejection: 3,180 grid rows failed the receive-age rule and 3,065 the quote-age rule on Hyperliquid, against 176 and 166 on Binance, and no row failed the validity, spread or gap rules. Hyperliquid had 504 receive gaps longer than 2 s in the range (the longest 173 s, on 26 September); Binance had 9. Feed quality improved late in the period: training days lost 9 to 21 windows each, while from 27 September onward only 1 to 4 windows per day were lost. One rejection per day is an artefact of the preparation code. Each day is cached separately, so the 00:00:00 grid row has no earlier record and the 00:00–00:05 window is always dropped; this accounts for 30 of the 395 rejections. Appendix E gives the counts by reason and day.

2.3 Venue asymmetry

Over the 30 days, Binance's 100 ms depth stream delivered 9.32 snapshots per second (the stream's cap is 10) and Hyperliquid's 12Book stream 1.84. The update rate alone does not show which venue moves first; Section 5.2 tests whether the gap predicts Hyperliquid's next move. At decision rows the median Hyperliquid quote age was 623, 627 and 640 ms on the training, validation and test splits, with a 99th percentile of 1.06 to 1.12 s. Part of that figure is the receive age, the time since the snapshot arrived, whose median was about 270 ms on every split. Receive time minus exchange timestamp had a median of 335, 339 and 350 ms and a 99th percentile of 0.60 to 0.65 s. This difference mixes network delay with any offset between the exchange clock and the recorder clock, so it is not a pure latency measurement. The proposal's figure of about 285 ms came from an earlier calculation whose definition we could not trace, so we do not compare with it. The median top-of-book spread was about 0.012 bps on Binance and 0.12 bps on Hyperliquid.

Because perpetual futures never expire, each venue keeps its contract price near the spot index price with funding: periodic payments between holders of long and short positions, at a rate the venue sets. A difference in funding between venues can therefore sustain a price offset between them. Funding payments, which the proposal planned to include in the reward, are not in the source table, and the funding rate is set to zero throughout (Section 8.1).

2.4 Gap distribution and the 18–23 September offset

As a check on the preparation, we recomputed the reference statistics in the team's data guide [5] for 15 September with the guide's own method (last snapshot in each one-second bucket, inner join of the two venues). We obtained a median absolute gap of 0.940 bps against the quoted 0.956, a 99th percentile of 4.36 against 4.30, and 0.218% of seconds above 7 bps, equal to the

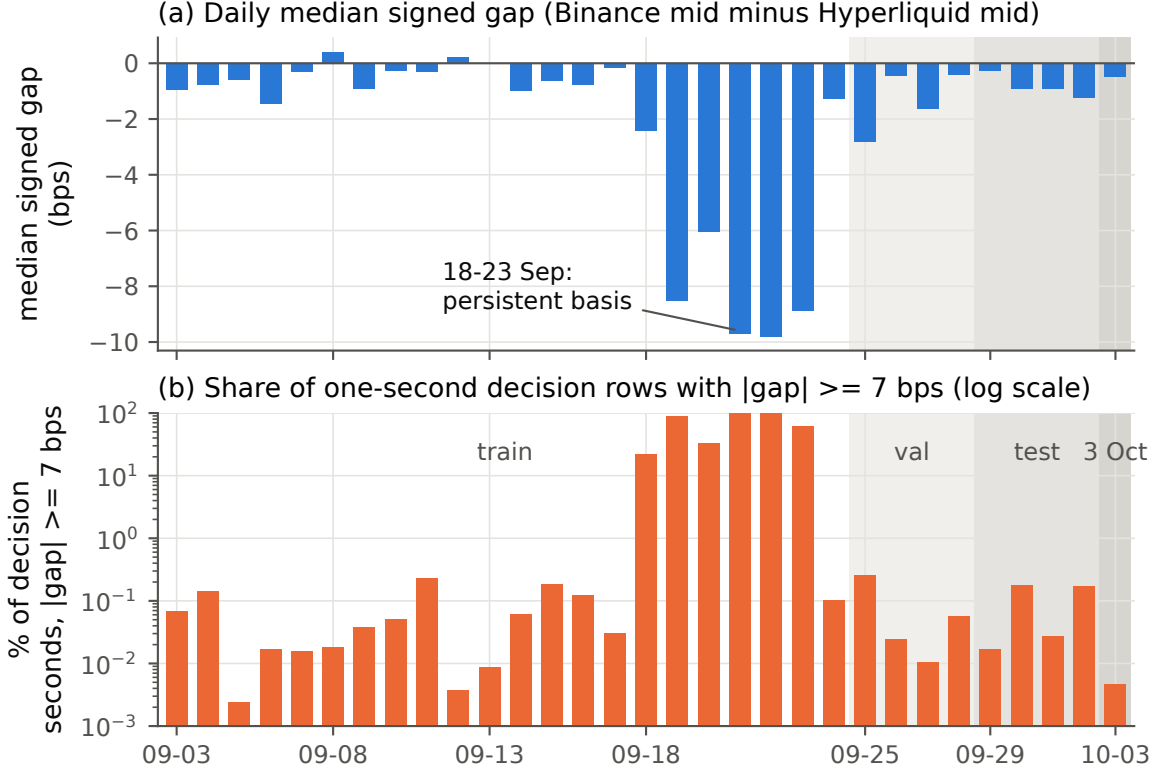


Figure 1: Daily cross-venue gap on prepared decision rows, 3 September to 3 October 2026. (a) Median signed gap. (b) Share of decision seconds with an absolute gap of at least 7 bps, roughly one round-trip cost, on a log scale. Shading marks the training (white), validation, test and fresh-day periods. Six training days carry almost all large gaps.

quoted value. Our causal episodes for the same day gave 0.931, 4.27 and 0.185%.

The training split has far more large gaps than the later splits (Table 1): on training days 18.04% of decision seconds had an absolute gap of at least 7 bps, against 0.084% on validation, 0.097% on test and 0.0047% (4 seconds) on 3 October. Figure 1 shows when they occurred. From 18 to 23 September the Binance mid sat below the Hyperliquid mid for whole days, with daily median signed gaps of -2.4 , -8.5 , -6.1 , -9.7 , -9.8 and -8.9 bps. These six days hold 322,900 of the 323,790 training seconds with an absolute gap of 7 bps or more (99.7%). Without them the training rate is 0.068%, close to validation and test. After we subtracted each window’s mean gap, only 0.012% to 0.152% of seconds on those days still deviated by 7 bps or more. The large gaps on these days were therefore a slow, persistent offset between the two prices (a basis). Once the offset was removed, deviations of 7 bps or more were about as rare on those days as on the other training days, so the days did not contain unusually many of the short-lived dislocations that latency arbitrage targets. We did not establish the cause of the offset (Section 8.7).

3 Environment and cost model

3.1 Decision process

The environment follows the Gymnasium interface [6]. Each episode replays one five-minute window and the agent acts once per second, so an episode has 300 decisions. The state is a vector of 17 numbers: the gap; the top-of-book spread on each venue; the order-book imbalance on each venue, $(\text{bid volume} - \text{ask volume}) / (\text{total volume})$ over 20 Binance and 5 Hyperliquid levels; Binance volatility (the standard deviation of mid-price log returns over the last 20 replay

rows, about 6.7 s); the Hyperliquid quote age and receive age; and account features (inventory, holding time, pending order type and remaining delay, drawdown from peak equity, equity profit and loss (P&L), time left, Hyperliquid mid price and unrealised P&L). The actions are HOLD, LONG, SHORT and EXIT on Hyperliquid. Recorded prices do not respond to the agent’s orders. An episode ends after its last decision and any open position is closed at the final row.

Equity is cash plus inventory times the Hyperliquid mid price. The reward is the change in equity between consecutive decisions, multiplied by a fixed scale (100 in training, 1 in evaluation). There is no bonus or penalty attached to any action. Fees, spread and slippage reach the reward through cash, so the unscaled rewards of an episode sum exactly to its net P&L; a unit test checks this.

Table 2: Settings of the base simulator. Attempt 2 kept these except the fee; its policy acted only at candidate seconds (about 755 decisions in 1,121 validation episodes), with a fixed exit rule.

Parameter	Value
Taker fee per side	3.5 bps (attempt 1); 4.5 bps (attempt 2)
Execution delay	150 ms (300 ms in stress tests)
Slippage per side	0.1 bps
Spread multiplier	1.0 (1.5 in stress tests)
Position size	0.001 BTC
Decision interval	1 s (300 decisions per episode)
Maximum holding time	30 s from entry fill
Entry freshness limit (quote and receive age)	1,000 ms
Drawdown stop	\$100
Forced-exit penalty beyond displayed depth	25 bps
Reference capital	\$10,000, reset each episode
Reward scale	100 (training), 1 (evaluation)
Funding	not available; set to zero

3.2 Execution and costs

An order decided at time t waits in a queue and fills at the first replay row at or after $t + 150$ ms. The delay stands for the time from seeing a Binance update to the order reaching Hyperliquid’s matching engine. The data guide [5] and the proposal set it at 150 ms, which the proposal calls conservative. We did not measure it: the receive-minus-exchange times in Section 2.3 include an unknown clock offset and describe the incoming feed rather than the order path, so they cannot calibrate it. We use 300 ms as a stress case. On the three-row grid the delayed fill uses the +150 ms row; all 32 fills of the threshold control on the test split were exactly 150 ms after their decisions. A fill walks the far side of the recorded Hyperliquid book (a buy takes asks, a sell takes bids) for the order size and pays the volume-weighted average price (VWAP) over the levels it consumes. Before the walk, each level’s distance from the mid is multiplied by a spread multiplier (1.0 by default and 1.5 in stress tests), and the VWAP is then moved 0.1 bps against the trader as slippage. A taker fee of 3.5 bps of the traded value is charged on every fill. A voluntary order larger than the displayed depth is rejected. An entry is also rejected if the Hyperliquid quote age or receive age exceeds 1,000 ms at the decision or at the fill, or if too little time remains to exit before the episode ends. The agent holds at most one position of 0.001 BTC, about \$80 at September prices (roughly \$75 to \$87), and cannot add to a position or reverse it in

one step. While an order is pending, only EXIT has an effect: it cancels a pending entry.

At a mid of \$84,000, the validation and test median, one round trip of 0.001 BTC pays $2 \times 3.5 \text{ bps} \times \$84 \approx \$0.059$ in fees, plus about 0.2 bps of slippage and 0.12 bps of spread, about 7.3 bps or \$0.061 in all. The fees recorded for the 16 test trades of the threshold control ranged from \$0.058 to \$0.061 per round trip. Table 2 lists the parameters.

3.3 Risk limits

A position is closed at the first row 30 s or more after its entry fill. A drawdown of \$100 from peak equity forces an exit and ends the episode. A forced exit that needs more than the displayed depth prices the remainder at the worst level plus a 25 bps penalty. With a 0.001 BTC position the drawdown stop never fired in any validation or test episode (losing \$100 would need a price move of more than 100%), and every test fill was covered by displayed depth. These two rules are safeguards for larger position sizes and did not constrain these experiments.

4 Methods

4.1 Controls

The threshold rule uses an entry threshold θ . When flat, it goes LONG on Hyperliquid if the gap is at least θ (Binance is above Hyperliquid, so Hyperliquid is expected to rise) and SHORT if the gap is at most $-\theta$. It exits a long when the gap falls to 0.5 bps or below and a short when the gap rises to -0.5 bps or above, and it does nothing while an order is pending. We chose θ on validation from $\{1, 2, 4, 7, 10, 20, 100\}$ bps by highest net P&L, then fewer trades, then larger θ . The flat control chooses HOLD at every step and earns exactly zero. Both controls run in the same environment and pay the same costs as PPO.

4.2 Attempt 1: four-action PPO

We used the Stable-Baselines3 [7] implementation of PPO [3], with separate policy and value networks of two 64-unit hidden layers, generalised advantage estimation [8] with $\lambda = 0.95$, discount $\gamma = 0.999$, clip range 0.2, learning rate 3×10^{-4} , 4,096 steps per rollout, minibatches of 256, 5 epochs per rollout, entropy coefficient 0.01, value coefficient 0.5 and gradient-norm clipping at 0.5 (Appendix B lists all settings). Each candidate trained for 1,003,520 steps (245 rollouts) on windows drawn uniformly from the training split. Observations were standardised with running statistics fitted on the training data and frozen for evaluation; rewards were not normalised. We trained three seeds (7, 17 and 27), since PPO results can vary widely between seeds [9]. The protocol, written and hashed before training, allowed one conditional retry: if all three seeds made zero validation trades, seed 7 would be retrained with the entropy coefficient raised to 0.05. That condition was met and the retry ran. Each candidate trained in about 15 to 17 minutes on one CPU thread. Evaluation used the most likely action at each step.

The PPO candidate was chosen by validation net P&L, then fewer trades, then the declared order of candidates. The selection file was written before the test split was read. The test (29 September to 2 October) was then run once for the selected PPO policy, the selected threshold rule and the flat control, followed by 16 cost scenarios replayed with the frozen policies: fee per side of 0.5, 1, 2 and 3.5 bps, delay of 150 and 300 ms, and spread multiplier of 1.0 and 1.5.

4.3 Attempt 2: opportunity-level PPO with a supervised warm start

Attempt 1 asked for a decision every second, and almost none of those seconds offered a trade worth making. Attempt 2 removed them. A candidate is a decision second at which the absolute

gap is at least 6 bps, both Hyperliquid ages are at most 1,000 ms, the displayed depth covers the order, and at least 31 s remain in the episode. At each candidate the agent chooses skip or enter. The trade direction follows the sign of the gap, and the exit is fixed: close when the gap returns to within 0.5 bps of zero, otherwise at the 30 s limit or the episode end. No candidate is offered while a position is open. Seconds between candidates are replayed without a decision, so one PPO step corresponds to one candidate. The state has 18 features: the absolute gap, the gap minus a round-trip cost estimate, both spreads, both imbalances, volatility, both Hyperliquid ages, the gap 1 s and 5 s earlier, each venue’s price change over the previous 1 s and 5 s (all signed in the trade direction, with flags for whether that history exists), and time left. The fee was raised to 4.5 bps per side, which we treated as Hyperliquid’s base taker rate; we did not check it against the venue’s published fee schedule.

Training had three stages. First, every training candidate was replayed under the fixed entry and exit rules to record its net counterfactual payoff in dollars. Second, the policy network was warm-started by behaviour cloning: binary cross-entropy with label 1 when the payoff was positive, each example weighted by its absolute payoff divided by the mean absolute payoff, for 50 epochs with Adam at learning rate 10^{-4} . Third, PPO fine-tuned the warm-started policy for 65,536 candidate decisions per seed (rollouts of 1,024, minibatches of 128, 5 epochs, learning rate 10^{-4} , entropy coefficient 0.005, $\gamma = 1$, two 32-unit layers, reward scale 100; Appendix B). We used seeds 7, 17 and 27 and kept both the warm-start and the PPO checkpoint of each seed as validation candidates, six in total, chosen by validation net P&L and then fewer trades. The test days of attempt 1 had already been seen, so they were excluded. The protocol would open 3 October as a fresh test only if the selected candidate had positive validation net P&L with at least one trade.

4.4 Metrics

We report net P&L (terminal equity minus \$10,000, summed over episodes without compounding), the number of completed trades, win rate (share of trades with positive net P&L), maximum drawdown in dollars, trade selectivity (trades divided by the number of decision seconds with an absolute gap of at least 7 bps, a count of large gaps rather than of profitable ones) and, for PPO, the share of decisions on which each action was chosen. All dollar amounts are for a 0.001 BTC position and are small in absolute terms, so we compare policies with each other under identical costs.

5 Results

5.1 Attempt 1: training

All four candidates followed the same path (Figure 2). The initial policy sampled the four actions about equally and closed about 49 trades per 300-step episode, losing about \$2.89 per episode, of which \$2.78 was fees. The share of LONG and SHORT among sampled actions first fell below 1% of a rollout at 176k, 209k, 229k and 184k steps for seeds 7, 17 and 27 and the retry. In the last tenth of training episodes 97% to 99% had no trade at all, and in the final rollout of every candidate LONG and SHORT were never sampled; the actions were split between HOLD and EXIT (2,513 and 1,583 for seed 7). Mean episode P&L was negative in every tenth of training for every seed, and over the whole run each candidate lost between \$368 and \$506, of which \$351 to \$488 was fees (Appendix C). The training callback recorded episode outcomes and action counts but not the value loss, entropy, approximate KL divergence or explained variance, so we cannot report those curves.

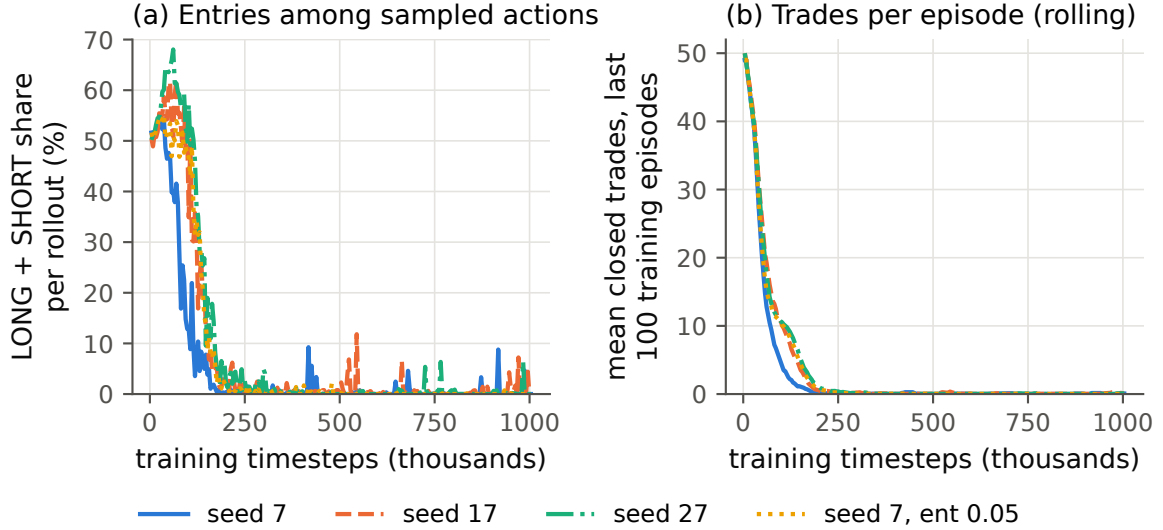


Figure 2: Attempt 1 training. (a) Share of LONG and SHORT among the 4,096 actions sampled in each rollout. (b) Mean closed trades per episode over the last 100 completed training episodes (a rolling window of about seven rollouts), recorded at the end of each rollout. All four candidates nearly stopped sampling entries within the first quarter of training, including the retry with five times the entropy coefficient.

5.2 Attempt 1: validation

Table 3: Attempt 1 on validation (25–28 September, 336,300 decisions, 284 seconds with $|\text{gap}| \geq 7$ bps). Fee 3.5 bps per side. Pre-fee P&L is after spread, slippage and delay.

Policy	Trades	Net P&L (\$)	Pre-fee P&L (\$)	Fees (\$)	Win rate
PPO seed 7 (selected)	0	0.00	0.00	0.00	—
PPO seed 17	0	0.00	0.00	0.00	—
PPO seed 27	0	0.00	0.00	0.00	—
PPO seed 7, entropy 0.05	0	0.00	0.00	0.00	—
Threshold 1 bps	11,895	−629.39	69.62	699.01	2.1%
Threshold 2 bps	6,708	−331.71	62.92	394.64	3.8%
Threshold 4 bps	2,014	−83.17	35.45	118.63	9.5%
Threshold 7 bps	144	−2.05	6.43	8.48	34.0%
Threshold 10 bps (selected)	14	0.14	0.96	0.82	42.9%
Threshold 20 and 100 bps	0	0.00	0.00	0.00	—
Flat	0	0.00	0.00	0.00	—

All four PPO candidates made zero trades on validation (Table 3). Their deterministic actions differed: seed 7 chose HOLD in 99.3% of decisions, while seeds 17 and 27 and the retry chose HOLD in 43.1%, 36.6% and 61.3%. Nearly all their other actions were EXIT while flat, which does nothing. Seed 17 and the retry issued 11 and 69 entry requests, all rejected by the freshness or end-of-episode checks. With every candidate at zero trades and \$0, seed 7 was selected by the declared order.

Net P&L rose with the threshold up to 10 bps: low thresholds traded thousands of times and lost heavily, and only the 10 bps rule finished positive, with \$0.14 over 14 trades. Before fees,

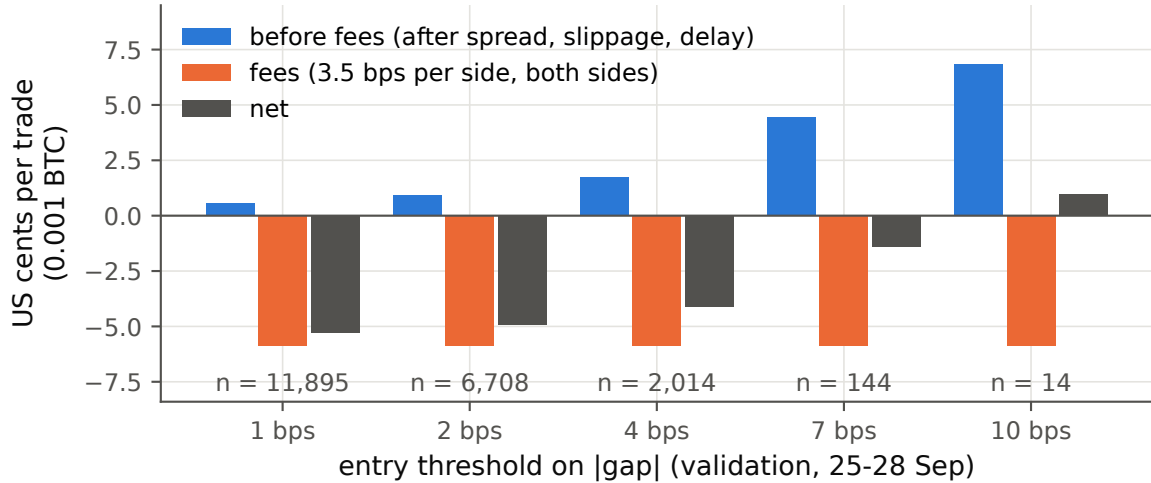


Figure 3: Attempt 1 validation, threshold rules: mean P&L per trade before fees, mean fee per trade, and mean net P&L per trade. n is the number of trades. The gross gain per trade grows with the threshold while the fee stays near 5.9 cents.

P&L was positive at every threshold that traded, so the gap does predict the direction of the next Hyperliquid move after spread, slippage and the 150 ms delay. The predicted move is small (Figure 3): the mean pre-fee gain per trade was 0.59 US cents at 1 bps and 6.86 cents at 10 bps, against a fee of about 5.9 cents per round trip. Only at 10 bps did the average trade cover its fees.

5.3 Attempt 1: held-out test

Table 4: Attempt 1 on the held-out test (29 September to 2 October, 1,140 episodes, 342,000 decisions, 331 seconds with $|\text{gap}| \geq 7$ bps), run once after selection. Fee 3.5 bps per side, 150 ms delay.

Policy	Trades	Net P&L (\$)	Fees (\$)	Win rate	Max DD (\$)	Selectivity
PPO (seed 7)	0	0.00	0.00	—	0.00	0.000
Flat	0	0.00	0.00	—	0.00	0.000
Threshold 10 bps	16	1.19	0.95	87.5%	0.10	0.048

PPO made no trades on the test days (Table 4). Its decision log has 338,980 HOLD and 3,020 EXIT-while-flat actions, no LONG or SHORT, and no entry requests, so its \$0 is the flat control's result. The 10 bps threshold control made 16 trades (8 long, 8 short) for a net \$1.19, which is 0.012% of the \$10,000 reference, after \$0.95 of fees. It had 14 winning trades, a maximum drawdown of \$0.10 and a mean holding time of 10.5 s; 12 exits came from the rule, 3 from the 30 s limit and 1 from the episode end. Of its 41 entry requests, 25 were rejected by the freshness or timing checks. The result is small and concentrated: 15 of the 16 trades and 98.7% of the net P&L fell on 30 September and 2 October, there was no trade on 1 October, and the three best trades produced 57% of the total.

This comparison depends on the fee. At 4.5 bps per side, the rate we later used for attempt 2, every threshold that traded would have lost money on validation (the 10 bps rule about \$0.10 over the same 14 trades, the 7 bps rule about \$4.47), so the declared selection rule would have chosen the 100 bps threshold, which never trades. The test comparison would then have been \$0 against \$0. These figures re-price the recorded validation trades at the higher fee, which is valid because the rule's trades do not change with the fee (Section 5.4); run directly at 4.5 bps in

attempt 2’s evaluation, the same rule made 13 trades and lost \$0.08 (Table 5). On the test days the 10 bps rule would still have made about \$0.92 at 4.5 bps, but it would not have been selected.

5.4 Attempt 1: cost sensitivity

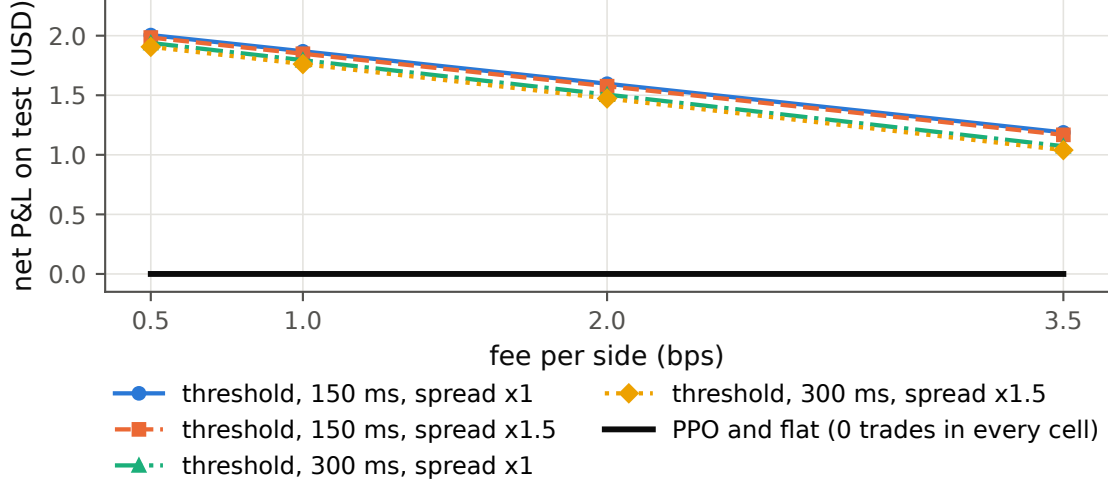


Figure 4: Attempt 1 test, net P&L against fee per side for the 10 bps threshold control under four delay and spread settings. The four lines nearly overlap; Table 8 in Appendix D gives the exact values. PPO and flat made no trades in any of the 16 cells.

Across the 16 cost scenarios, PPO and flat made zero trades in every cell (Figure 4). The threshold control stayed positive in every cell, from \$2.01 (0.5 bps, 150 ms, spread $\times 1$) down to \$1.04 (3.5 bps, 300 ms, spread $\times 1.5$). The 300 ms delay changed its trade count from 16 to 17 and lowered its win rate at 3.5 bps from 87.5% to 76.5%. The sampled grid stops at 3.5 bps per side, so no simulated cell brackets the fee at which the control breaks even. Because the rule’s decisions do not depend on the fee, its trades are the same at every fee level and its net P&L falls linearly with the fee rate. Solving for zero gives a break-even fee of 7.1 to 7.9 bps per side across the four delay and spread settings, an extrapolation from 16 or 17 trades.

5.5 Attempt 2: opportunity-level PPO

The training split produced 317,232 candidates from 1,971 of the 5,984 training windows. Of their counterfactual payoffs at 4.5 bps per side, 4,270 (1.35%) were positive, 308,663 (97.30%) negative and 4,299 zero (entries the environment rejected). The mean payoff was -7.66 cents. The warm-start loss fell from about 0.32 to about 0.035 after two epochs (0.035 to 0.036 across seeds) and to 0.025 after 50. PPO fine-tuning entered on 4 to 7 of the 1,024 candidates in each seed’s first rollout and on 111, 106 and 73 of 65,536 candidates in total; seeds 17 and 27 entered on none in their last rollout. Training P&L was $-\$4.58$, $-\$5.31$ and $-\$4.17$ over 102, 99 and 66 trades.

On validation the four days offered about 755 candidates in 1,121 episodes, and 934 episodes had none. All three PPO checkpoints and the seed-7 warm start made zero trades (Table 5). The seed-17 and seed-27 warm starts made four trades each and lost money. The best threshold rule at 4.5 bps per side was 12 bps, with \$0.16 over 2 trades. The selected candidate was the seed-7 warm start, chosen by tie-break among the zero-trade candidates. It did not meet the fresh-test gate, so attempt 2 never opened 3 October. The outcome file for this run records the result as a failure to learn profitable trading.

Table 5: Attempt 2 on validation (25–28 September), fee 4.5 bps per side. Warm-start checkpoints are supervised models before PPO fine-tuning.

Candidate or control	Trades	Net P&L (\$)	Note
Seed 7 warm start (selected)	0	0.000	1 entry, rejected
Seed 7 PPO	0	0.000	
Seed 17 warm start	4	−0.108	1 winner
Seed 17 PPO	0	0.000	
Seed 27 warm start	4	−0.021	2 winners
Seed 27 PPO	0	0.000	
Threshold 6 bps	338	−15.102	
Threshold 7 bps	130	−3.965	
Threshold 9 bps	21	−0.302	
Threshold 10 bps	13	−0.079	
Threshold 12 bps	2	0.159	best rule
Threshold 15 bps	1	0.142	
Threshold 20 bps	0	0.000	
Flat	0	0.000	

6 Why PPO learned not to trade

We identify five properties of the problem and the training set-up that plausibly contributed to the failure; we did not test them separately. We found no fault in the environment or evaluation code: their tests pass, unscaled rewards reconcile with the trade ledger, and the threshold control trades profitably in the same environment. The PPO inference path (saved model, frozen normalisation and action decoding) is covered only by the saved-model and normalisation equivalence tests.

6.1 Staying flat returns zero, which random entry cannot beat

With a round trip costing about 7.3 bps and a median absolute gap of 1.3 bps, almost every entry has a negative expected value. The initial random policy lost \$2.89 per episode. Starting from that policy, lowering the probability of entering raises the expected return, and the flat policy’s return of zero is higher than that of any policy that enters at random. It is not the best policy available: the 10 bps rule beat it on validation and test. Once entries became rare, few entry samples remained to update their probability; we did not log advantages, so we cannot confirm this mechanism. Figure 2 shows entries becoming rare within the first quarter of training for every seed.

6.2 Trades that cover their costs are rare

On validation, 284 of 336,300 decision seconds (0.084%) had an absolute gap of 7 bps or more and 31 had 10 bps or more. The average threshold trade covered its fees only at 10 bps, where 14 trades in four days netted about 1 cent each (Figure 3). To learn to enter at those seconds and nowhere else, the agent must pick out that gain from a reward stream in which a wrongly timed entry costs about 6 cents.

Large gaps often coincide with stale Hyperliquid quotes. On the test days the quote age or receive age exceeded 1,000 ms in 2.5% of all decision seconds, but in 31% of seconds with an absolute gap of 7 bps or more and in 49% of those with 10 bps or more (on validation, 1.7%,

21% and 39%). The entry-freshness rule refuses entries at exactly those seconds; with the timing check, it rejected 25 of the threshold control’s 41 test entry requests. Part of the measured gap is therefore likely the age of the Hyperliquid snapshot rather than a tradable price difference. The rule is also applied to ages that include clock offset (Section 2.3).

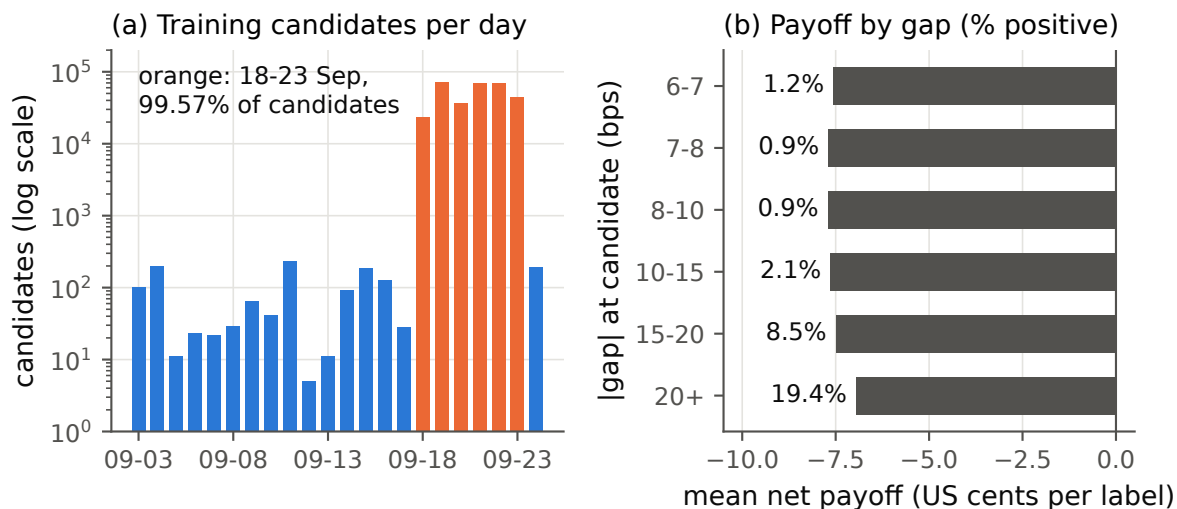


Figure 5: Attempt 2 training labels at 4.5 bps per side. (a) Candidates per training day on a log scale; the six orange days (18–23 September) hold 99.57% of them. (b) Mean counterfactual net payoff by absolute gap at the candidate, with the share of positive labels printed beside each bar. The mean is negative in every bucket.

Attempt 2’s training labels were also negative on average in every gap bucket, including gaps of 20 bps or more, where 19.4% were positive (Figure 5b and Appendix F). But 99.57% of them came from the six basis days (Section 6.4), so they say little about the other days.

6.3 Attempt 1 had two ways of doing nothing

While the agent is flat, EXIT has the same effect as HOLD. In the final training rollouts the sampled actions were split between the two, and three of the four deterministic validation policies chose EXIT in 39% to 63% of decisions. A policy split evenly between HOLD and EXIT keeps half of the maximum entropy ($\ln 2$ of $\ln 4$ nats) without entering. The retry with five times the entropy coefficient nearly stopped entering at about the same point as the original seed 7 (184k against 176k steps) and made no validation trades. We read this as evidence that the redundant action made abstention easier to reach in attempt 1. Attempt 2 had a single way to abstain (skip), and its PPO seeds also made zero validation trades, so the redundant action is at most a contributing factor and was not required for abstention. We did not run an ablation that removes it from attempt 1.

6.4 The training period was dominated by a basis regime

Of the training seconds with an absolute gap of 7 bps or more, 99.7% fell on 18–23 September, and so did 99.57% of attempt 2’s candidates (Figure 5a). Only 17 of those 315,861 candidates had a positive signed gap. On the six basis days 99.72% of the counterfactual trades ran 30 s or more from decision to completion, nearly all exactly 31 s, which is how a close at the 30 s holding limit shows in the replay; on the other 16 training days the share was 27.0% (median 8 s). Only 1.27% of the labels on the six days were positive (mean -7.69 cents), against 19.9% on the other 16 days, where the mean label was still -1.81 cents. In attempt 1 these six days supplied 1,623 of the 5,984 training windows (27.1%); in attempt 2 about 99.5% of PPO decisions came

from them. In attempt 2 the labels on these days show that large gaps rarely closed within 30 s; we did not test whether attempt 1 learned this.

Offsets were not confined to the training period. The first validation day, 25 September, had a daily median gap of -2.8 bps, larger in size than on 18 September (-2.4 bps), although only 0.25% of its seconds reached 7 bps against 22% on 18 September. After we subtracted each window's mean gap, 83 validation seconds and 127 test seconds still deviated by 7 bps or more, against 284 and 331 before; on the 16 non-basis training days the count fell only from 890 to 643. Slow offsets therefore produced most of the large gaps on validation and test as well.

6.5 The warm start learned to skip because almost all training labels lost money

Skip was the loss-minimising answer on these labels: 97.3% were negative, and positives were 1.35% of the examples but carried 0.51% of the weight under the $|\text{payoff}|$ weighting. The labels also overlap in time. Most are consecutive seconds within the same windows of the same six days, and their counterfactual trades could not all be taken, so the effective number of independent examples is far below 317,232. The weighted loss fell from about 0.32 to about 0.035 within two epochs and moved little after that. PPO then started from a policy that almost never entered, and entries made up only 0.11% to 0.17% of its decisions.

6.6 What the threshold control does and does not show

The \$1.19 on the test belongs to the fixed 10 bps rule, not to PPO. It shows that the environment allows trades that cover their costs at 3.5 bps per side and that the rule found 16 of them on the test days. It does not show a dependable edge: the profit rests on 16 trades concentrated on two days, and the rule was selected only because validation was priced at 3.5 bps (Section 5.3).

7 Responsible AI

7.1 Fairness

Latency arbitrage earns money by trading against quotes that have not caught up with the market. The other side of those trades is the liquidity provider whose quote is stale, often a slower participant. Budish et al. [1] argue that this race is a cost to liquidity providers that is passed on to all traders as wider spreads. A working version of our agent would be on the fast side of that transfer. Our results suggest that, after taker fees, the amount available to a small taker on this pair is small, but the concern applies to any team with lower fees or faster infrastructure. Nothing in this project connects to a live market or uses capital.

7.2 Transparency

The experiment protocols were written and hashed before training, every prepared episode and model file has a SHA-256 hash, and the test runs logged every decision, fill and trade. Failed experiments were kept. The two PPO runs each have an outcome file that records the failure and states that no success claim is permitted; the diagnostic in Appendix G has only a status file, whose conclusion calls its single fresh-day trade net positive. Each table and figure in this report names the split it comes from, and the statistics we computed for this report beyond the saved run summaries are reproduced by a script stored with it (Appendix A).

7.3 Accountability

Policies were selected on validation data only, and the test was run once, after the selection file was frozen. The test days are now recorded as spent. Risk limits are enforced by the environment

rather than left to the learned policy. The AI-use statement below records which parts were produced by AI agents, and the team members are responsible for reviewing them.

7.4 Trust and calibrated claims

The main risk of misuse of these results is over-reading them: describing a zero-trade policy as one that “avoided losses”, or describing the threshold control’s profit as a reinforcement learning result (Section 6.6). We report zero trades as abstention and give the trade count next to every P&L figure, so that results resting on a handful of trades are visible as such.

8 Limitations and threats to validity

8.1 Funding is assumed zero

The source table has no funding rates, so funding is set to zero in every result. Positions last at most 30 s, so the direct effect on single trades is probably small, but we could not check it, and a funding difference between the venues is one possible cause of the 18–23 September basis.

8.2 Quality-filtered subset

We kept 95.43% of the five-minute windows. Dropping windows with stale Hyperliquid quotes removes some of the moments when the slower venue lags most, which may bias the data toward calmer periods. Keeping only windows in which every row passed the quality checks also conditions each episode on the absence of a feed outage later in the same five minutes, which a live trader would not know. One window per day is dropped for a preparation reason unrelated to quality (Section 2.2).

8.3 Sampled books

Both feeds are snapshot streams, and Hyperliquid’s has 5 levels and about 1.84 updates per second. Between snapshots the true book is unknown, and a fill at +150 ms uses the latest recorded Hyperliquid snapshot, which can be close to a second old. The quote age we report includes a median of about 270 ms since the snapshot was received, and the rest mixes network delay with clock offset between machines.

8.4 No market impact or competition

Orders take displayed liquidity without moving later prices, and other traders who react to the same Binance move are not modelled. The delay is fixed at 150 ms (300 ms in stress tests) rather than drawn from a measured distribution.

8.5 Small trade counts

The threshold control made 14 trades on validation and 16 on test, and the attempt-2 candidates made between 0 and 4. These counts are too small to estimate a profit rate with useful precision.

8.6 Reuse of evaluation data

The test days of attempt 1 have been seen and cannot serve as fresh evidence. Attempt 2 was designed after attempt 1 failed and was selected on the same validation days, and the fee setting moved from 3.5 to 4.5 bps between attempts. The fresh day of 3 October was later opened by a non-RL diagnostic (Appendix G), so no untouched day remains in the current dataset.

8.7 Instrument identity and the basis

The rows carry no symbol field. The Binance stream name `btcusdt@depth20@100ms` does not by itself record whether the spot or the futures endpoint was used, and the two venues may quote Bitcoin against different stablecoins. A quote-currency difference, a premium or funding difference between the venues, or a venue-specific event could each explain the 18–23 September basis; we did not verify any of them.

8.8 Limited search

Each attempt used one hyperparameter configuration (plus the one entropy retry in attempt 1) and a fixed training budget. We did not try reward shaping, an action set without the redundant EXIT, a value-based agent such as DQN [10], which the proposal listed as an optional comparison, or a longer training run. The observation includes the absolute Hyperliquid price, which drifts over time and relies on the training-set normalisation. Optimisation diagnostics (value loss, entropy, KL divergence) were not logged. The study covers one pair of venues for one month.

9 Conclusion

We built a causal replay of 30 days of Binance and Hyperliquid order books, a cost-aware environment with delayed fills against the recorded book, and two pre-declared PPO experiments with validation-only selection. Neither PPO design learned to trade. The four-action agent cut its share of entry actions below 1% within about 230,000 of its 1,000,000 training steps on every seed, made zero trades on validation, and made zero trades on the held-out test, where a fixed 10 bps threshold rule made \$1.19 over 16 trades at 3.5 bps per side. At 4.5 bps per side the declared selection rule would not have chosen that rule. The opportunity-level agent with a supervised warm start also made zero validation trades on all three PPO seeds. The project did not reach its stated goal of beating the fixed rule under identical costs. Under these costs staying flat returned zero while random entry lost money, cost-covering trades were rare, and the training period’s large gaps came mostly from a six-day basis that did not close within the holding limit. Attempt 1’s redundant EXIT action may have made abstention easier to reach, but attempt 2 abstained without it.

Lessons and next steps

We changed the set-up once (attempt 2) and the algorithm not at all, so we cannot say which matters more. Next steps for both:

- Attempt 2 removed the redundant action, among other changes, and still abstained, so changing the action set is not enough on its own.
- Separate large gaps that come from a slow offset between venues from gaps that come from one venue lagging, for example by measuring the gap relative to a slowly moving estimate of the basis. Offsets produced most large gaps on all three splits, not only in training.
- Merge labels from consecutive seconds of the same gap event into one opportunity.
- Keep reporting the flat and threshold controls next to the agent, report results at more than one fee, require more trades before any claim of profit, and use newly recorded days for the next test, since every day in the current dataset has now been examined.
- Log PPO’s optimisation statistics (value loss, entropy, KL divergence and advantages) so that a collapse can be diagnosed while it happens, and try the untested options in Section 8.8, such as a value-based agent or a broader hyperparameter search.

Team roles

Roles to be confirmed by each member. The proposal planned the following division; the AI-use statement below records which parts AI agents implemented.

- *Kerk Tai Heng*: data preparation and episode generation.
- *Member 2*: replay environment and transaction-cost model.
- *Member 3*: baseline strategy and evaluation tools.
- *Member 4*: PPO training and tuning.
- *Member 5*: held-out evaluation and fee sensitivity tests.
- *Member 6*: Responsible-AI analysis, report integration and presentation.

TODO (team): Replace the placeholders and edit each line so that it states what each member actually did. Adjustments from the proposal are allowed but should be stated.

AI-use statement

AI coding agents did a large share of the engineering and writing in this project, under the team's direction. An OpenAI Codex-based coding agent implemented substantial parts of the pipeline: the data preparation and quality checks, the replay environment and cost model, the threshold and flat controls, the evaluation code, the PPO training and experiment drivers for both attempts, the gradient-boosting diagnostic in Appendix G, and the run-report scripts. Anthropic's Claude audited the saved run artifacts against the source code, recomputed the statistics marked as derived in this report, wrote the figure script (`make_figures.py`), and drafted the text of this report. The research question, the data recording, the cost assumptions and the evaluation plan come from the team's proposal. The team reviewed the agents' output and is responsible for the code, the results and this report. ChatGPT was used to edit the proposal for clarity.

TODO (team): Confirm or edit this statement so that it matches exactly what each member and each AI tool did, including which parts of the code and text the team wrote or rewrote, before submission.

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Appendices

A Reproducibility and provenance

All runs used Python 3.12.3, NumPy 2.5.3, PyTorch 2.14.1 (CPU), Gymnasium 1.3.0 and Stable-Baselines3 2.9.0; the diagnostic in Appendix G also used scikit-learn 1.9.1 [11]. Training ran on a CPU: about 65 minutes for the four attempt-1 candidates and about 24 minutes for the whole attempt-2 pipeline. The saved artifacts are:

- Prepared data: the 30-day set in `data/recorded-30d/`, whose manifest has SHA-256 `afa9c485...46b1`, and the 3 October holdout in `data/recorded-fresh-2026-10-03/`.
- Attempt 1: `runs/recorded-ppo-2026-10-04/` (git commit `98cbef6`, protocol SHA-256 `84d1e980...6f40`, selection SHA-256 `c23f9219...9f07`).
- Attempt 2: `runs/opportunity-ppo-2026-10-04/`; an independent audit before selection confirmed that source and manifest hashes matched and that no fresh-day output existed.
- Appendix G diagnostic: `runs/opportunity-value-2026-10-04/`.

The figures and the statistics this report computed beyond the saved run summaries are produced by `docs/report_v1/make_figures.py`, which reads only saved artifacts and writes `docs/report_v1/generated/derived_stats.json`. They are the daily gap statistics, per-decile training statistics, per-trade threshold economics, the threshold rules re-priced at 4.5 bps per side, the attempt-2 label statistics, and, per split, the Hyperliquid quote and receive ages, spreads, prices, snapshot rates and stale-quote shares. While checking the artifacts we found three discrepancies in older project documents: the data guide’s update rates (8.24 and 1.63 per second) do not match its own rows per day, the proposal’s “roughly 26 million time-aligned, ten-level order-book snapshots” does not match the 28.9 million raw rows with five Hyperliquid levels, and the 3 October data manifest still says no trading evaluation was performed on that day although the run in Appendix G later evaluated it.

B PPO hyperparameters

Table 6: PPO settings for both attempts. Attempt-2 values for clip range, λ , value coefficient and gradient clip are the library defaults and were recorded with the run. An attempt-2 training step is one candidate decision.

Setting	Attempt 1	Attempt 2
Action space	HOLD, LONG, SHORT, EXIT	skip, enter
Observation features	17	18
Training steps	1,003,520	65,536
Steps per rollout	4,096	1,024
Minibatch size	256	128
Epochs per rollout	5	5
Learning rate	3×10^{-4}	10^{-4}
Discount γ	0.999	1.0
GAE λ	0.95	0.95
Clip range	0.2	0.2
Entropy coefficient	0.01 (retry 0.05)	0.005
Value coefficient	0.5	0.5
Gradient-norm clip	0.5	0.5
Hidden layers (policy and value)	64, 64	32, 32
Observation clip	10	10
Reward scale in training	100	100
Seeds	7, 17, 27 (+ retry 7)	7, 17, 27
Warm start	none	50 epochs

C Attempt 1 training diagnostics per candidate

Table 7: Attempt 1 training, from the per-episode monitor logs (3,345 training episodes per candidate). “First decile” and “last decile” are the first and last tenth of training episodes. “Entry share below 1%” is the training step at the end of the first rollout in which LONG and SHORT made up less than 1% of sampled actions. “Retry” is seed 7 with entropy coefficient 0.05.

	Seed 7	Seed 17	Seed 27	Retry
Training time (s)	1,017	984	896	978
Total trades	6,262	8,568	8,742	8,100
Total net P&L (\$)	−368.24	−495.97	−506.13	−472.74
Total fees (\$)	351.32	476.75	487.78	453.77
Episodes with no trade	84.5%	79.4%	78.5%	79.4%
Trades per episode, first decile	17.47	22.08	21.44	20.38
Trades per episode, last decile	0.066	0.105	0.042	0.006
No-trade episodes, last decile	99.1%	97.0%	98.2%	99.4%
Final rollout HOLD	2,513	2,055	2,021	2,375
Final rollout EXIT	1,583	2,041	2,075	1,721
Entry share below 1% at step	176,128	208,896	229,376	184,320
Validation HOLD share	99.3%	43.1%	36.6%	61.3%
Validation entry requests	0	11	0	69

Training rollouts sample from the stochastic policy, while validation and test use the most likely action. This is why a few training episodes late in training still contain trades while the deterministic policies never trade.

D Attempt 1 full test sensitivity grid

Table 8: Threshold control (10 bps) on the test days under each cost scenario. PPO and flat made 0 trades and \$0 in all 16 scenarios. Break-even is the fee per side at which net P&L reaches zero, computed from the fixed trade set (not simulated).

Delay (ms)	Spread mult.	Fee 0.5 (\$)	Fee 1.0 (\$)	Fee 2.0 (\$)	Fee 3.5 (\$)	Break-even (bps/side)
150	1.0	2.005	1.869	1.597	1.188	7.86
150	1.5	1.985	1.848	1.576	1.168	7.79
300	1.0	1.939	1.794	1.506	1.072	7.21
300	1.5	1.906	1.762	1.473	1.040	7.10

Trades were 16 at 150 ms and 17 at 300 ms in every fee setting. Win rates at 3.5 bps were 87.5% (150 ms) and 76.5% (300 ms); at 0.5 bps every trade won.

E Data-quality detail

Table 9: Grid rows (of 7,776,000) that failed a quality rule, 3 September to 2 October. A failed row rejects its whole five-minute window.

Reason	Binance	Hyperliquid
Receive age over 2,000 ms	176	3,180
Quote age over 3,000 ms	166	3,065
No earlier snapshot	30	103
Latest snapshot invalid	0	0
Gap over 250 bps (either venue)	0	
Receive gaps over 2 s (count)	9	504
Longest receive gap	31.7 s (21 Sep)	173.0 s (26 Sep)

Rejected windows per day were 9 to 21 on the 22 training days (most on 15 September); 18, 10, 1 and 2 on the validation days; and 4, 2, 3 and 3 on the test days. Of the 504 Hyperliquid receive gaps, 455 were in the training period, 35 in validation and 14 in test. On 3 October 286 of 288 windows were kept and Hyperliquid had 3 gaps over 2 s (longest 109.4 s). Every episode file holds 900 rows with the fields listed in the data schema, and in all 8,531 episodes the decision rows are rows 0, 3, 6, ... and fall on whole seconds.

F Attempt 2 label statistics

Table 10: Attempt 2 counterfactual labels by absolute gap at the candidate (training split, 4.5 bps per side). Payoffs are in US cents for 0.001 BTC.

gap (bps)	Candidates	Positive share	Mean net payoff (cents)
6–7	32,879	1.2%	−7.56
7–8	50,466	0.9%	−7.69
8–10	141,738	0.9%	−7.70
10–15	89,218	2.1%	−7.64
15–20	2,281	8.5%	−7.49
20 or more	650	19.4%	−6.95
All	317,232	1.35%	−7.66

The median payoff was −7.59 cents; positive labels averaged +2.94 cents and negative labels −7.92 cents, and the mean fee per traded label was 7.53 cents. On the six basis days 1.27% of labels were positive (mean −7.69 cents); on the other 16 training days, 1,371 candidates in all, 19.9% were positive (mean −1.81 cents). PPO fine-tuning lost \$4.58, \$5.31 and \$4.17 on training replay (seeds 7, 17, 27), paying \$7.54, \$7.32 and \$4.96 in fees.

G Non-RL diagnostic: expected-value selector (not PPO)

After attempt 2, the coding agent ran a supervised diagnostic that is not reinforcement learning and is not part of the PPO results. Two gradient-boosted regression models (scikit-learn

HistGradientBoostingRegressor, 200 iterations) were fitted to predict the counterfactual net payoff of the 317,232 attempt-2 training candidates. Each model was paired with a minimum predicted value of \$0 or \$0.01 for entering, giving four candidates; the six attempt-2 checkpoints were re-evaluated unchanged alongside them. Validation used 3.5 bps per side, so the fee changed again. The regressors were fitted to payoffs labelled at 4.5 bps per side. During replay at 3.5 bps the cost-excess input was lowered by 2 bps to match its training value, so the entry rule compared a payoff predicted at 4.5 bps per side with the margin. The gate for opening 3 October required only positive validation net P&L with at least one trade; it did not require beating the threshold rule.

On validation the selected model (15 leaves, minimum 50 samples per leaf, \$0 margin) made \$0.058 over 2 trades, while the 12 bps threshold rule made \$0.193 over 2 trades and the 10 bps rule \$0.139 over 13. All three PPO checkpoints again made zero trades. The selected model passed the gate, so 3 October was opened. That day offered only 4 candidates in 286 episodes. The model and the 12 bps threshold rule skipped the same three and took the same one, a 14.4 bps gap at 21:05:18 UTC: a short of 0.001 BTC held for 5 s that netted \$0.045 at 3.5 bps per side and \$0.028 at 4.5 bps. The two policies' decision, trade and fill logs are byte-identical. One trade identical to the threshold rule's is no evidence of a learned edge, and running this diagnostic used up the last untouched day of data.